Controlled comparison of four imbalanced-learning methods

UGO-SMOTE • BFCH-SMOTE • EC-EBO • SSHR

Supervisor briefing | Ecoli4 and Yeast4 | 13 October 2026

Purpose and principal finding

This descriptive pilot compares four implemented resampling approaches under the same held-out folds. No resampling and ordinary SMOTE serve as controls. UGO-SMOTE had the highest observed mean minority F1 on Ecoli4 with both classifiers; EC-EBO had the highest observed mean F1 on Yeast4 with both classifiers. There is no method that leads on every dataset and metric.

Adding unlabeled training features to SSHR reduced mean F1 and raised false-positive rate in all four dataset/classifier conditions. Some Yeast4 balance and ranking metrics improved, so the result is a tradeoff rather than a universal judgment about semi-supervised learning.

Design and inputs

Dataset	Rows	Features	Majority / minority	Imbalance ratio
Ecoli4	336	7	316 / 20	15.8 : 1
Yeast4	1,484	8	1,433 / 51	28.1 : 1

Protocol: five repeats of five stratified-group outer folds (25 fold events), root seed 42. Identical raw feature-coordinate groups remain together across train/test and labelled/unlabeled training pools. Approximately 50% of training labels are retained. Every method receives the same observed labelled pool L; only SSHR receives the remaining training-only feature pool U. Hidden labels are unavailable to SSHR. Preprocessing is fitted on L and the held-out test fold is used only for final scoring.

Classifiers: logistic regression (LogReg) and Gaussian naive Bayes (GNB). Main methods: no resampling, SMOTE, UGO-SMOTE, BFCH-SMOTE, EC-EBO, and SSHR. UGO-SMOTE is the UGO variant in this common comparison; UGO-ROS is not included. The empty-U SSHR control uses the same folds, masks, observed labels and classifier settings.

At-a-glance: mean minority F1

Method	Ecoli4 LogReg	Ecoli4 GNB	Yeast4 LogReg	Yeast4 GNB
No resampling	0.872	0.769	0.202	0.277
SMOTE	0.819	0.736	0.276	0.230
UGO-SMOTE	**0.889**	**0.773**	0.252	0.264
BFCH-SMOTE	0.795	0.720	0.302	0.224
EC-EBO	0.725	0.587	**0.319**	**0.279**
SSHR	0.584	0.571	0.296	0.272

The bold cells mark the largest observed mean in each condition; they are not statistical superiority claims. Full mean ± sample SD tables for all five metrics follow.

Ecoli4: complete held-out metric comparison

336 rows; 316 majority and 20 minority. All entries are mean ± sample standard deviation over the same 25 successful fold events. FPR is a proportion and lower is preferable.

LogReg

Method	F1 ↑	G-mean ↑	MCC ↑	ROC-AUC ↑	FPR ↓
No resampling	0.8720 ± 0.1085	0.9009 ± 0.0929	0.8738 ± 0.1056	0.9955 ± 0.0071	0.0032 ± 0.0065
SMOTE	0.8194 ± 0.1350	0.9381 ± 0.0732	0.8158 ± 0.1397	0.9951 ± 0.0070	0.0210 ± 0.0199
UGO-SMOTE	**0.8890 ± 0.1071**	0.9287 ± 0.0863	0.8885 ± 0.1076	0.9951 ± 0.0072	0.0051 ± 0.0075
BFCH-SMOTE	0.7949 ± 0.1398	0.9492 ± 0.0579	0.7971 ± 0.1369	0.9960 ± 0.0058	0.0292 ± 0.0270
EC-EBO	0.7247 ± 0.1218	0.9373 ± 0.0586	0.7279 ± 0.1182	0.9940 ± 0.0100	0.0424 ± 0.0282
SSHR	0.5838 ± 0.1640	0.9331 ± 0.0342	0.6130 ± 0.1464	0.9916 ± 0.0130	0.1006 ± 0.0560

GNB

Method	F1 ↑	G-mean ↑	MCC ↑	ROC-AUC ↑	FPR ↓
No resampling	0.7694 ± 0.1663	0.8661 ± 0.1214	0.7670 ± 0.1713	0.9871 ± 0.0160	0.0152 ± 0.0168
SMOTE	0.7363 ± 0.1511	0.8788 ± 0.1104	0.7348 ± 0.1543	0.9888 ± 0.0134	0.0260 ± 0.0251
UGO-SMOTE	**0.7729 ± 0.1679**	0.8664 ± 0.1218	0.7704 ± 0.1728	0.9871 ± 0.0160	0.0146 ± 0.0165
BFCH-SMOTE	0.7201 ± 0.1471	0.8521 ± 0.1165	0.7141 ± 0.1517	0.9850 ± 0.0145	0.0215 ± 0.0188
EC-EBO	0.5871 ± 0.1392	0.9093 ± 0.0748	0.6016 ± 0.1355	0.9838 ± 0.0254	0.0822 ± 0.0446
SSHR	0.5714 ± 0.2103	0.8580 ± 0.1589	0.5824 ± 0.2103	0.9757 ± 0.0264	0.0779 ± 0.0612

Ecoli4 interpretation: UGO-SMOTE has the highest mean F1 (0.8890 LogReg; 0.7729 GNB). For LogReg, the no-resampling FPR is 0.0032, compared with 0.1006 for SSHR. BFCH-SMOTE leads Ecoli4/LogReg in G-mean (0.9492), but its FPR (0.0292) is also higher than no resampling. A classifier FPR change does not by itself measure geometric class overlap.

Yeast4: complete held-out metric comparison

1,484 rows; 1,433 majority and 51 minority. All entries are mean ± sample standard deviation over the same 25 successful fold events. FPR is a proportion and lower is preferable.

LogReg

Method	F1 ↑	G-mean ↑	MCC ↑	ROC-AUC ↑	FPR ↓
No resampling	0.2018 ± 0.1759	0.3050 ± 0.2242	0.2331 ± 0.1889	0.8718 ± 0.0652	0.0046 ± 0.0047
SMOTE	0.2757 ± 0.0811	0.7898 ± 0.0757	0.3083 ± 0.0798	0.8698 ± 0.0574	0.1258 ± 0.0290
UGO-SMOTE	0.2519 ± 0.1808	0.3961 ± 0.2717	0.2717 ± 0.1796	0.8737 ± 0.0615	0.0106 ± 0.0104
BFCH-SMOTE	0.3025 ± 0.0875	0.7739 ± 0.0807	0.3234 ± 0.0892	0.8545 ± 0.0682	0.0995 ± 0.0253
EC-EBO	**0.3189 ± 0.0852**	0.7854 ± 0.0746	0.3414 ± 0.0825	0.8705 ± 0.0667	0.0923 ± 0.0186
SSHR	0.2962 ± 0.0908	0.7935 ± 0.1284	0.3363 ± 0.0881	0.8781 ± 0.0729	0.1107 ± 0.0351

GNB

Method	F1 ↑	G-mean ↑	MCC ↑	ROC-AUC ↑	FPR ↓
No resampling	0.2768 ± 0.1243	0.5895 ± 0.1255	0.2580 ± 0.1268	0.8357 ± 0.0854	0.0488 ± 0.0171
SMOTE	0.2301 ± 0.0642	0.7598 ± 0.0857	0.2599 ± 0.0711	0.8187 ± 0.0847	0.1535 ± 0.0382
UGO-SMOTE	0.2639 ± 0.1220	0.5680 ± 0.1313	0.2438 ± 0.1254	0.8326 ± 0.0836	0.0470 ± 0.0177
BFCH-SMOTE	0.2239 ± 0.0779	0.6872 ± 0.1179	0.2312 ± 0.0917	0.7782 ± 0.0847	0.1246 ± 0.0469
EC-EBO	**0.2789 ± 0.0807**	0.7782 ± 0.0918	0.3065 ± 0.0881	0.8360 ± 0.0969	0.1150 ± 0.0246
SSHR	0.2722 ± 0.0810	0.7032 ± 0.1294	0.2825 ± 0.0821	0.8356 ± 0.0872	0.0933 ± 0.0477

Yeast4 interpretation: EC-EBO has the highest observed mean F1 (0.3189 LogReg; 0.2789 GNB). For LogReg, its F1 exceeds no resampling (0.2018), while FPR rises from 0.0046 to 0.0923. The GNB F1 margin over no resampling is only 0.0021. These differences are descriptive; the study does not establish statistical significance.

Effect of unlabeled data within SSHR

The paired ablation compares SSHR with U against the same SSHR implementation with an empty U pool. The signed difference below is with U minus without U on 25 matching fold events. Positive FPR means more false alarms.

Condition	ΔF1	ΔG-mean	ΔMCC	ΔROC-AUC	ΔFPR
Ecoli4 / LogReg	-0.0777	-0.0160	-0.0678	-0.0028	+0.0303
Ecoli4 / GNB	-0.0922	-0.0054	-0.0810	-0.0078	+0.0361
Yeast4 / LogReg	-0.0096	+0.0172	-0.0018	-0.0071	+0.0145
Yeast4 / GNB	-0.0054	+0.0810	+0.0017	+0.0296	+0.0333

F1 falls and FPR rises in all four conditions. Yeast4/LogReg G-mean rises by 0.0172; Yeast4/GNB G-mean and ROC-AUC rise by 0.0810 and 0.0296. These gains occur alongside lower F1 and higher FPR. The ablation measures the total effect of access to U, including clustering and selection changes, rather than isolating one internal mechanism.

What these results answer

Boundary question: BFCH-SMOTE is operational and sometimes raises G-mean, but the reported FPR tradeoffs do not demonstrate that generated samples avoid true class overlap. Geometric diagnostics would be needed for that specific claim.

Intra-class imbalance question: EC-EBO leads mean F1 on Yeast4 but not Ecoli4. Overall classifier scores do not by themselves show that every minority subcluster was improved.

Unlabeled-data question: SSHR uses existing U observations with minority pseudo-labels; only its SMOTE fallback creates new feature vectors. In this two-dataset pilot, access to U provides no consistent F1 benefit. The committed retrospective audit reports pseudo-label correctness of 170/313 Ecoli4 and 96/273 Yeast4 selection events per classifier. Repeated fold events are not unique source observations.

Limits, evidence, and discussion with supervisor

This is a two-dataset, two-classifier pilot under our declared common protocol, not the original papers' full benchmark or their exact experimental setup. Author-code equivalence and the original SSHR unlabeled-pool protocol remain unverified. The repeated folds share observations; sample SDs are descriptive, not confidence intervals or independent replications. No statistical or general superiority claim follows from these tables.

Evidence: the user-exported all_methods_comparison.csv.xlsx contains 120 unique main-result rows, all with 25 folds, and the pasted SSHR ablation contains 20 unique rows, all with 25 paired folds. Both were cross-checked against the committed pilot CSVs (numeric agreement within Excel rounding). Repository: MAYANK-RATHVA05/rb_ugo, sshr-implementation at 5813e57; study record: docs/SSHR_SMALL_STUDY.md; full aggregate CSV/JSON evidence: docs/sshr_pilot/. Raw feature files and individual job JSON records are outside this report.

Suggested discussion: Which research question should drive the next controlled experiment—reducing boundary overlap, improving minority subcluster coverage, or filtering low-confidence SSHR pseudo-labels? Any next dataset or parameter policy should be fixed before inspecting its held-out scores.